

CS & IT ENGINEERING



Computer Network

IPv4 Address

Lecture No. - 04



By - Abhishek Sir



Recap of Previous Lecture



Topic

Network Mask

(Netmask)





Topics to be Covered



Topic

Classful IPv4 Address

Topic

Subnetting

ABOUT ME



Hello, I'm **Abhishek**

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[MCQ]

[GATE-2010][2 Mark]



#Q. Suppose computers A and B have IP addresses "10 . 105 . 1 . 113" and "10 . 105 . 1 . 91" respectively and they both use the same netmask N. Which of the values of N given below should not be used if A and B should belong to the same network?

☒ A 255 . 255 . 255 . 0 $\Rightarrow$ 24 one's

☐ B 255 . 255 . 255 . 128 $\Rightarrow$ 25 one's

☒ C 255 . 255 . 255 . 192 $\Rightarrow$ 26 one's

☒ D 255 . 255 . 255 . 224 $\Rightarrow$ 27 one's

Ans: D

①

10.105. 1.113
255.255. 255.224

10.105. 1.96

10.105. 1.91
255.255. 255.224

10.105. 1.64



113 → 01110001
224 → 11100000
96 ← 01100000

91 → 01011011
224 → 11100000
64 ← 01000000

②

10.105. 1.113
255.255. 255.192

10.105. 1.64

10.105. 1.91
255.255. 255.192

10.105. 1.64

113 → 01110001
192 → 11000000
64 ← 01000000

91 → 01011011
192 → 11000000
64 ← 01000000

C

10.105. 1.113
255.255. 255.128

10.105. 1. 91
255.255. 255.128

10.105. 1. 0

10.105. 1. 0

113 → 01110001
128 → 10000000
0 ← 00000000

A

10.105. 1.113
255.255. 255. 0

10.105. 1. 91
255.255. 255. 0

10.105. 1. 0

10.105. 1. 0

91 → 01011011
128 → 10000000
0 ← 00000000

10.105.1.113 = 10.105.1.01110001

10.105.1.91 = 10.105.1.01011011

255.255.255.0 = 255.255.255.00000000

255.255.255.128 = 255.255.255.10000000

255.255.255.192 = 255.255.255.11000000

255.255.255.224 = 255.255.255.11100000

27 bit NetID

[MCQ]

[GATE-2003][2 Mark]



#Q. The subnet mask for a particular network is "255 . 255 . 31 . 0". Which of the following pairs of IP addresses could belong to this network ?

- ☒ A 172 . 57 . 88 . 62 and 172 . 56 . 87 . 233
- ☒ B 10 . 35 . 28 . 2 and 10 . 35 . 29 . 4
- ☒ C 191 . 203 . 31 . 87 and 191 . 234 . 31 . 88
- ☒ D 128 . 8 . 129 . 43 and 128 . 8 . 161 . 55

Ans: D



172. 57. 88. 62
255. 255. 31. 0

172. 57 . . 0

172. 56. 87. 233
255. 255. 31. 0

172. 56 . . 0

28 → 00011100
31 → 00011111

28 ← 00011100

10. 35. 28. 2
255. 255. 31. 0

10. 35. 28 . 0

10. 35. 29. 4
255. 255. 31. 0

10. 35. 29 . 0

29 → 00011101
31 → 00011111

29 ← 00011101

191.203.31.87

255.255.31.0

191.203. . 0

191.234.31.88

255.255.31.0

191.234. . 0

129 → 10000001
31 → 00011111
1 ← 00000001

128.8.129.43

255.255.31.0

128.8. | . 0

128.8.161.55

255.255.31.0

128.8. | . 0

161 → 10100001
31 → 00011111
1 ← 00000001



Topic : IPv4 Address



Problem : How many bits are assigned for network id field ?

→ IPv4 address are categorized into two type :

1. Classful IPv4 Address

→ Static Assignment : NetID field size defined implicitly



2. Classless IPv4 Address

→ Dynamic Assignment : Need to define NetID field size explicitly



Topic : Classful IPv4 Address

→ Classful IPv4 Address are of five types :

1. Class A

2. Class B

3. Class C

4. Class D

5. Class E

→ [NetID and HostID]

→ [All are special IP
cannot be used as host IP]



Topic : Class A IPv4 Address

→ IP Address = 0.....
[Start with Zero bit]

→ IP address range :

0 . 0 . 0 . 0

to

127 . 255 . 255 . 255

↓
[0 0000000 00000000 00000000 00000000]

31 bits

↑
[0 11111111 11111111 11111111 11111111]



Topic : Class A IPv4 Address



Network ID : 8 bits

Host ID : 24 bits

Max^m Number of Networks

$$= 2^7 = 128 / 126$$

Max^m Number of Hosts per Network

$$= (2^{24} - 2) \text{ hosts}$$

→ Default Network Mask for Class A IPv4 network :

255.0.0.0



Topic : Class B IPv4 Address

→ IP Address = 10.....
[Start with One & Zero bit]

→ IP address range :

128.0.0.0

to

191.255.255.255

[10000000 00000000 00000000 00000000]

30 bits

[10111111 11111111 11111111 11111111]

30 bits



Topic : Class B IPv4 Address



Network ID : 16 bits

Host ID : 16 bits ✓

Max^m Number of Networks = 2^{14}

Max^m Number of Hosts per Network = $(2^{16} - 2)$ hosts

→ Default Network Mask for Class B IPv4 network :

255.255.0.0



Topic : Class C IPv4 Address

→ IP Address = 110.....
[Start with One, One & Zero bit]

→ IP address range :

192.0.0.0

to

223.255.255.255

[11000000 00000000 00000000 00000000]

29 bits

[11011111 11111111 11111111 11111111]

29 bits

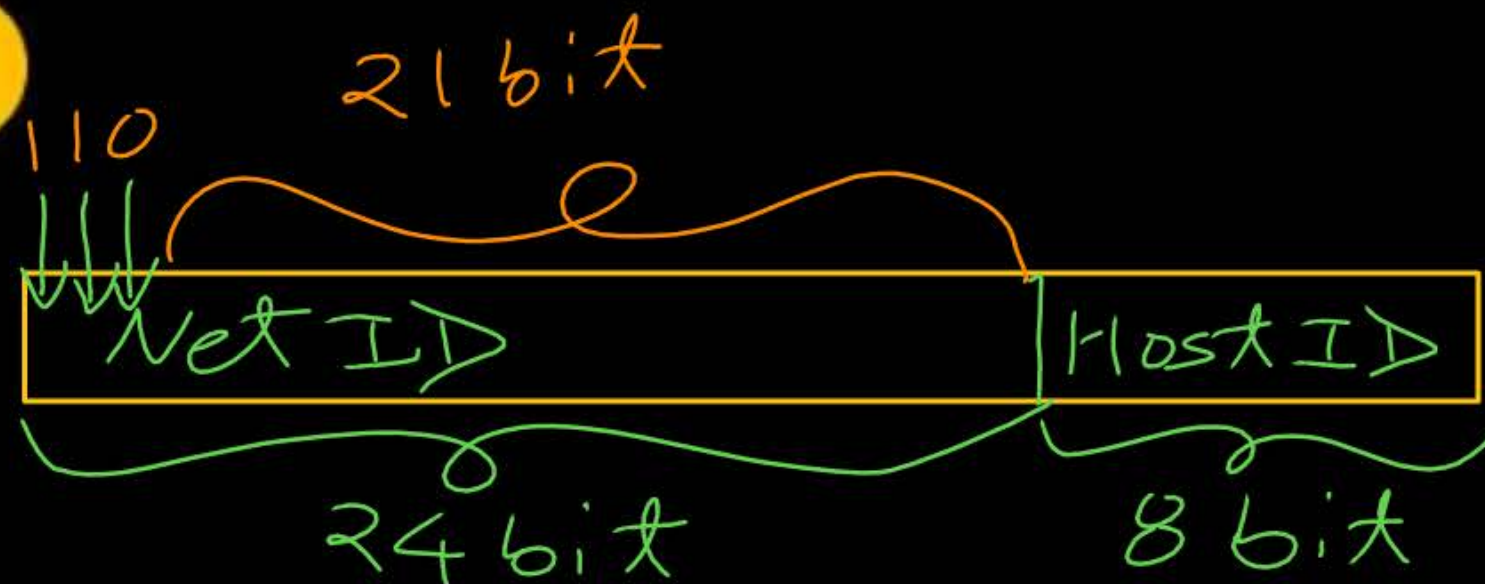


Topic : Class C IPv4 Address



Network ID : 24 bits

Host ID : 8 bits



Max^m Number of Networks = 2^{21}

Max^m Number of Hosts per Network = $(2^8 - 2)$ hosts

→ Default Network Mask for Class C IPv4 network : 255.255.255.0

[MCQ]

[1 IT - 1] [GATE-2012][1 Mark]



#Q. In the IPv4 addressing format, the number of networks allowed under Class C addresses is :

Class C :-

Net ID $\Rightarrow$ 24 bits

$$\text{Max}^m \text{ no. of networks} = \cancel{2^{24}} = 2^{21}$$

A 2^7

B 2^{14}

✓ C 2^{21}

D 2^{24}

Ans: C



Topic : Class D IPv4 Address

→ IP Address = 1110.....
[Start with One, One, One & Zero bit]

→ IP address range :

224.0.0.0

to

239.255.255.255

[11100000 00000000 00000000 00000000]

28 bits

[11101111 11111111 11111111 11111111]

28 bits



Topic : Class D IPv4 Address

→ Used for "Multicasting"

→ Class D IPv4 address can not be a host IP address

(Special IP Add.)

→ It is used as "Group Id" by IGMP
[Internet Group Message Protocol]



Topic : Class E IPv4 Address

→ IP Address = 1111.....
[Start with four One bit]

→ IP address range :

240 . 0 . 0 . 0

to

255 . 255 . 255 . 255

[11110000 00000000 00000000 00000000]

28 bit

[11111111 11111111 11111111 11111111]

28 bit

→ Reserved for future use



Topic : Classful IPv4 Address



classful
class A
class B
class C

Class	IP	NetID Bits	First Octet	Efficency
Class A	<u>0</u>	<u>8 - bits</u>	<u>0 to 127</u>	[50 %]
Class B	<u>10</u>	<u>16 - bits</u>	<u>128 to 191</u>	[25 %]
Class C	<u>110</u>	<u>24 - bits</u>	<u>192 to 223</u>	[12.5 %]
Class D	<u>1110</u>	[N. A]	<u>224 to 239</u>	[6.25 %]
Class E	<u>1111</u>	[N. A]	<u>240 to 255</u>	[6.25 %]

100% (2^{32})



Topic : Classless IPv4 Address ✓



→ Number of bits for network id field are assigned 'Dynamically'

→ No any prefix bits for IPv4 address
[Unlike Classful IPv4 Address]

$$0 \leq P \leq 255$$

→ Classless IPv4 Address : P.Q.R.S / x

→ Prefix or Slash notation [x] represent "number of network id bits"



Topic : Subnetting



- Dividing (logically) a network into smaller manageable sub-networks
- Sub-network (subnet) : Clustering of hosts inside a network
- Clustering of hosts based on some bits of host identifier (HostId) field
[In practice, most significant bits of host identifier]



Topic : Subnetting

Suppose,

Net ID field size is 27 bits

Host ID field size is 5 bits

→ one-bit
subnetting

2 subnet

Host ID

0	0	0	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
0	0	1	0	0
0	0	1	0	1
0	0	1	1	0
0	0	1	1	1
0	1	0	0	0
0	1	0	0	1
0	1	0	1	0
0	1	0	1	1
0	1	1	0	0
0	1	1	0	1
0	1	1	1	0
0	1	1	1	1

1	0	0	0	0
1	0	0	0	1
1	0	0	1	0
1	0	0	1	1
1	0	1	0	0
1	0	1	0	1
1	0	1	1	0
1	0	1	1	1
1	1	0	0	0
1	1	0	0	1
1	1	0	1	0
1	1	0	1	1
1	1	1	0	0
1	1	1	0	1
1	1	1	1	0
1	1	1	1	1





Topic : Subnetting

Suppose,

Net ID field size is 27 bits

Host ID field size is 5 bits

→ Two-bit
Subnetting

4 subnets

Host ID

0	0	0	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
0	0	1	0	0
0	0	1	0	1
0	0	1	1	0
0	0	1	1	1

0	1	0	0	0
0	1	0	0	1
0	1	0	1	0
0	1	0	1	1
0	1	1	0	0
0	1	1	0	1
0	1	1	1	0
0	1	1	1	1

1	0	0	0	0
1	0	0	0	1
1	0	0	1	0
1	0	0	1	1
1	0	1	0	0
1	0	1	0	1
1	0	1	1	0
1	0	1	1	1

1	1	0	0	0
1	1	0	0	1
1	1	0	1	0
1	1	0	1	1
1	1	1	0	0
1	1	1	0	1
1	1	1	1	0
1	1	1	1	1





Topic : Subnetting

Suppose,

Net ID field size is 27 bits

Host ID field size is 5 bits

→ 3-bit subnetting

$$2^3 = 8 \text{ subnets}$$

Host ID

0 0 0 0 0

0 0 0 0 1

0 0 0 1 0

0 0 0 1 1

0 0 1 0 0

0 0 1 0 1

0 0 1 1 0

0 0 1 1 1

0 1 0 0 0

0 1 0 0 1

0 1 0 1 0

0 1 0 1 1

0 1 1 0 0

0 1 1 0 1

0 1 1 1 0

0 1 1 1 1

1 0 0 0 0

1 0 0 0 1

1 0 0 1 0

1 0 0 1 1

1 0 1 0 0

1 0 1 0 1

1 0 1 1 0

1 0 1 1 1

1 1 0 0 0

1 1 0 0 1

1 1 0 1 0

1 1 0 1 1

1 1 1 0 0

1 1 1 0 1

1 1 1 1 0

1 1 1 1 1



2 mins Summary



Topic

Classful IPv4 Address ✓

Topic

Subnetting



THANK - YOU

